

Jianmin Liao

MACHINE LEARNING • COMPUTER VISION • COMPUTATIONAL IMAGING

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Summary

Ph.D. candidate developing data-efficient machine learning for image restoration and inverse problems, with a first-author paper accepted to NeurIPS 2026 (Main Track).

Education

Ph.D.	Mathematics (Statistics emphasis) , Syracuse University	Aug 2022–May 2027 (Expected)
M.S.	Computational Mathematics , Sun Yat-sen University	Sep 2019–Jun 2022
B.S.	Information and Computing Science , Sun Yat-sen University	Sep 2015–Jun 2019

Research Experience & Publications

Oracle Supervision Transfers for Hyperparameter Prediction in Model-Based Image Denoising Jan 2026–Present

First and corresponding author; NeurIPS 2026 Main Track (accepted)

- Conceived HyperDn, designed the method and experiments, and wrote the manuscript; the configuration-conditioned PyTorch predictor uses a 2.18M-parameter ConvNeXtV2 backbone and transfers TV/TGV supervision from 13 source configurations to diffusion-based DiffPIR.
- With **2 DiffPIR target oracle labels**, achieved **30.23 dB**—within 0.90 dB of oracle and 2.50 dB above the identical architecture without source pretraining on WED 256×256 .
- Without target oracle labels, transferred to two mixtures of individually seen noise types and three evaluated resolutions (321×481 , 512×768 , $\sim 1024 \times 768$), with 0.01–0.27 dB gaps to oracle.
- For L2-TV Gaussian, selected hyperparameters in 14–47 ms versus 2.04–10.86 s for Monte Carlo SURE on three OOD datasets, excluding the shared downstream solver.
- Extended transfer to SSIM and LPIPS objectives: with metric-specific training, **one target label** outperformed a from-scratch baseline using **16 labels** on both metrics.
- Transferred to unseen NLM and CBM3D denoisers with different hyperparameter semantics, achieving the best mean PSNR among evaluated baselines at all seven tested few-shot budgets.

MG-SpaIR: Multi-Grade Sparse-Guided Implicit Representation for Training-Data-Free Image Restoration Mar 2025–Jul 2026

Co-first and corresponding author; J. Math. Imaging Vis. 68, Article 48 (2026)

- Co-developed a PyTorch framework that restores images from a single observation without an external training set under combined blur, downsampling, noise, and missing pixels.
- On the paper's mixed-degradation evaluation, achieved the highest average PSNR/SSIM among compared training-data-free approaches; multi-grade training used **30% less VRAM** than single-grade.
- Across SIREN, WIRE, and Instant-NGP, validated the regularizer and established per-grade objective convergence and asymptotic stationarity under explicit regularity and step-size conditions.

A Successive Two-stage Method for Sparse Generalized Eigenvalue Problems Jul 2019–Aug 2023

Second author; arXiv preprint 2308.11904 (2023)

- Implemented and benchmarked a MATLAB support-alteration solver for sparse PCA, Fisher discriminant analysis, and canonical correlation analysis on problems with up to 20,000 variables.

Technical Skills

Programming	Python, C++, SQL, MATLAB
ML / Scientific	PyTorch, NumPy, pandas, scikit-learn
Methods / Tools	Deep learning, computer vision, image restoration, transfer learning, diffusion-based restoration, implicit neural representations, numerical optimization, Linux, Git, HTCondor

Additional: Teaching Assistant, Syracuse University & Sun Yat-sen University (2019–Present); Reviewer, IPCO (2024); Coursework & Qualifying Exams Prize, Syracuse University (2024–2025)